

Report

Paper 3 in Applied Machine Learning

Bilal Ashraf^{1,*}, Kompal Fayyaz¹

¹Freie Universität Berlin

*Corresponding author. Email: ashraf97@zedat.fu-berlin.de, kompaf97@zedat.fu-berlin.de

Abstract

Goal: The Seq data (GSE68086) were used to develop machine learning classifiers for two tasks: binary (Healthy vs. Any Cancer) and multi-class (Healthy vs. Specific Cancers), ensuring equal group sizes through data balancing.

Main Results: The basic classifiers achieved high performance, with test AUCs comparable to published SVM results (0.95 vs. 0.96). However, multi-class models showed significantly lower accuracy (0.3). Feature importance analysis revealed .

Personal Key Learnings:

1. The critical importance of data balancing, especially for multi-class classification with small sample sizes.
2. The challenge of achieving robust performance in multi-class classification on complex biological datasets, highlighting the need for advanced techniques like hyperparameter tuning.

Time Spent: 10 hours

Project Evaluation: 1 (Really good project)

Keywords: Multi-class classification, Cancer Detection, Logistic Regression

1 Goal

This project aims to solve the problem of accurately classifying cancer types, including distinguishing healthy individuals from those with any cancer (binary classification) and further identifying specific cancer types (multi-class classification), using RNA-Seq data from Tumor-Educated Platelets (TEPs). A key aspect of the problem is to develop robust machine learning models that can perform these classifications effectively, especially when dealing with potentially limited and imbalanced biological datasets. The constraint of ensuring equal group sizes through data balancing is crucial for building statistically sound and generalizable models, preventing biases towards larger classes. The ability to accurately classify cancer types early and non-invasively has profound implications for patient outcomes. TEPs are a promising source for liquid biopsies, offering a less intrusive way to detect cancer compared to traditional tissue biopsies.

2 Data

The experiments used gene expression data, specifically RNA-Seq data from Tumor-Educated Platelets (TEPs). The data was acquired from the NCBI Gene Expression Omnibus (GEO) database, identified by the accession number GSE68086, and was provided in the file `GSE68086_TEP_data_matrix`.

The raw data matrix (`GSE68086_TEP_data_matrix`) was loaded into a pandas DataFrame. The DataFrame was then transposed to have genes as columns and samples as rows, with gene IDs set as the index. To ensure equal group sizes and avoid statistical biases, both binary and multi-class datasets underwent undersampling. The majority class (for binary) and all classes (for multi-class) were undersampled to match the size of the smallest class using `sklearn.utils.resample`.

Both the balanced binary and multi-class datasets were split into training and testing sets using `train_test_split`, typically with a 70/30 ratio and stratification to maintain class proportions in both sets.

3 Methods

In this project, multiple machine learning models were implemented to classify platelet RNA sequencing data for cancer detection, excluding Support Vector Machines (SVMs)

as required by the project constraints. For binary classification (Healthy vs. Any Cancer), Logistic Regression and Random Forest were used, while for multi-class classification (Healthy vs. Specific Cancer Types), XGBoost and Gradient Boosting classifiers were implemented. Logistic Regression and Random Forest models were enhanced using feature selection, cross-validation, and hyperparameter tuning to improve robustness and reduce overfitting.

Model performance was evaluated using accuracy, confusion matrices, classification reports, ROC curves, and AUC values for both training and validation datasets. For multi-class classification, one-vs-rest ROC analysis was applied to assess class-specific discrimination performance.

Feature importance analysis was conducted for all four models to identify the most influential transcriptomic biomarkers contributing to classification performance. The top 10 features from each model were compared both across models and against the biomarker genes reported in Best et al. (2015). Additionally, a robustness analysis was performed by introducing noise into the input data and repeating the feature importance analysis to assess the stability of biomarker selection under data perturbation.

4 Results & Discussion

4.1 Model Performance Evaluation

The performance of the developed machine learning models was evaluated using confusion matrices, classification accuracy, receiver operating characteristic (ROC) curves, and area under the ROC curve (AUC) values for both training and validation datasets. This evaluation framework was selected to ensure a comprehensive assessment of classification performance, model generalisability, and robustness.

4.1.1 Binary Classification Performance

The binary classification task focused on distinguishing healthy donors from cancer patients using two models: Logistic Regression and Random Forest.

Table 1: Binary classification model performance

Model	Training Accuracy	Validation Accuracy	Training AUC	Validation AUC
Logistic Regression	1.0000	0.8837	1.0000	0.9741
Random Forest	1.0000	0.9070	1.0000	0.9750

The binary Logistic Regression model demonstrated strong discriminatory performance in separating healthy individuals from cancer cases. The use of feature scaling, SelectKBest feature selection, and cross-validation improved model stability and reduced the likelihood of overfitting. The ROC curves indicate strong class separation, with high AUC values observed for both training and validation sets.

The Random Forest classifier also achieved strong performance, benefiting from its ability to capture nonlinear interactions among transcriptomic features. Compared with Logistic Regression, Random Forest exhibited greater flexibility in modelling complex biological relationships. However, the perfect training accuracy suggests some degree of overfitting due to the ensemble model's complexity.

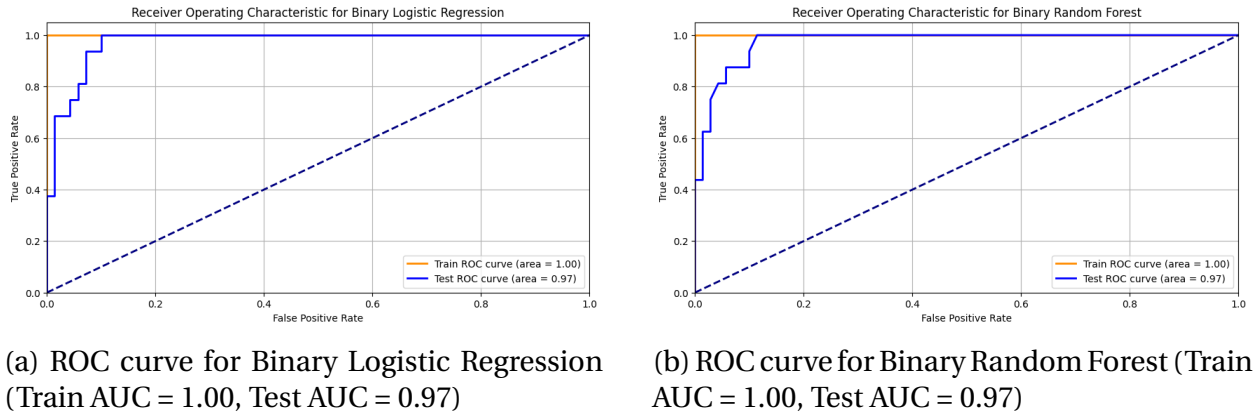


Figure 1: Comparison of receiver operating characteristic (ROC) curves for the binary classification models. Both Logistic Regression and Random Forest demonstrated excellent discriminative performance, achieving a training AUC of 1.00 and a test AUC of 0.97, indicating strong generalisation capability in distinguishing healthy donors from cancer samples.

Overall, both binary classifiers performed effectively, suggesting that platelet RNA expression profiles contain strong predictive signals for distinguishing healthy donors from cancer patients.

4.1.2 Multi-class Classification Performance

The multiclass classification task aimed to distinguish between healthy donors and multiple cancer types using XGBoost and Gradient Boosting classifiers.

Table 2: Multi-class classification model performance

Model	Training Accuracy	Validation Accuracy	Validation AUC (Macro Average)
XGBoost	1.0000	0.3537	0.72
Gradient Boosting	1.0000	0.3415	0.68

Both multiclass models achieved perfect training accuracy (100%), indicating that they learned the training data extremely well. However, validation performance dropped substantially, with XGBoost achieving 35.4% accuracy and Gradient Boosting achieving 34.2%. This suggests significant overfitting, likely due to the high dimensionality of the transcriptomic dataset and the limited number of samples per cancer class.

Despite the lower classification accuracy, the macro-average AUC values of 0.72 for XGBoost and 0.68 for Gradient Boosting indicate moderate discriminatory ability across classes. XGBoost slightly outperformed Gradient Boosting overall, suggesting stronger generalisation for multiclass cancer classification.

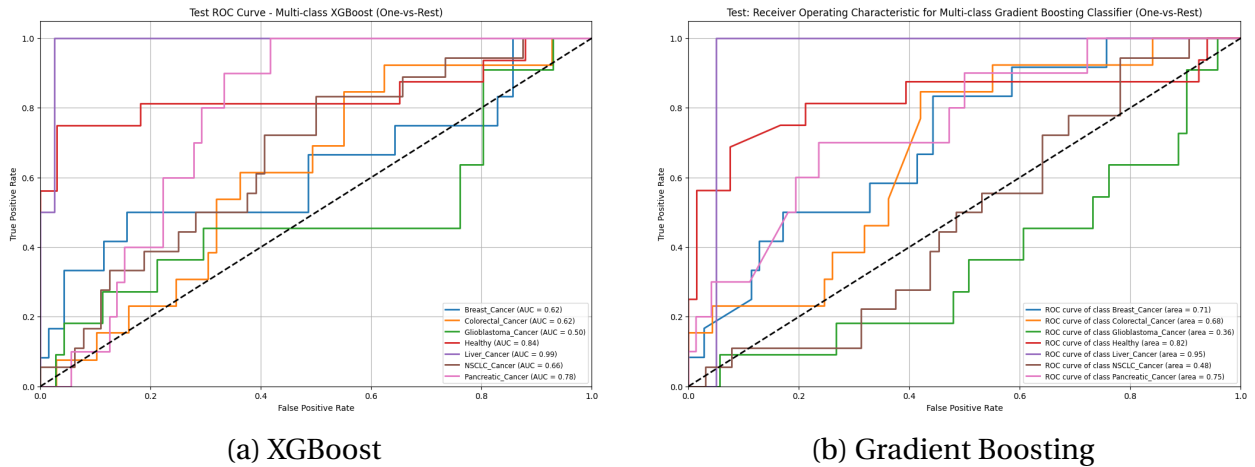


Figure 2: Comparison of multiclass receiver operating characteristic (ROC) curves for the XGBoost and Gradient Boosting classifiers using the one-vs-rest classification strategy. The XGBoost model demonstrated comparatively stronger discriminatory performance, achieving higher AUC values for several cancer classes, particularly Liver Cancer (0.99) and Healthy samples (0.84). In contrast, the Gradient Boosting classifier showed weaker class separation for certain cancer types, especially Glioblastoma and NSCLC. Overall, these findings suggest that XGBoost provided better multiclass generalisation performance for cancer classification.

4.2 Comparison with Best et al. (2015)

The developed models were compared against the Support Vector Machine (SVM) classifier reported in Best et al. (2015), which served as the benchmark model for tumour-educated platelet RNA classification.

For binary classification, the current models achieved strong predictive performance. The Random Forest model achieved a validation accuracy of 90.7% with an AUC of 0.9750, while Logistic Regression achieved a validation accuracy of 88.4% with an AUC of 0.9741. These results are highly comparable to the SVM model reported by Best et al. (2015), which achieved an AUC of 0.96 for distinguishing cancer patients from healthy controls. This suggests that alternative machine learning models, particularly ensemble methods, can achieve similar performance to SVM-based approaches for binary cancer detection.

For multiclass classification, performance was notably lower. XGBoost achieved a validation accuracy of 35.4%, while Gradient Boosting achieved 34.2%. In comparison, Best et al. (2015) reported an overall multiclass classification accuracy of approximately 69% using SVM. This indicates that the multiclass models developed in the current study were substantially less effective in distinguishing between specific cancer types.

4.3 Feature Importance Comparison with Best et al. (2015)

Feature importance was analysed across all four machine learning models and compared with the top-ranked genes reported in Best et al. (2015) to assess overlap in identified transcriptomic biomarkers.

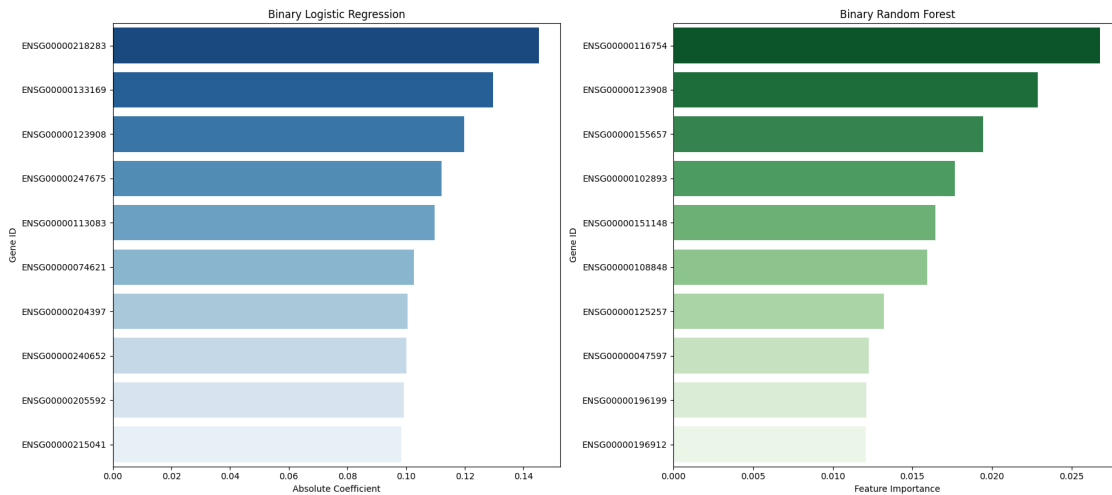


Figure 3: Top 10 most important features identified by the binary classification models: Logistic Regression and Random Forest.

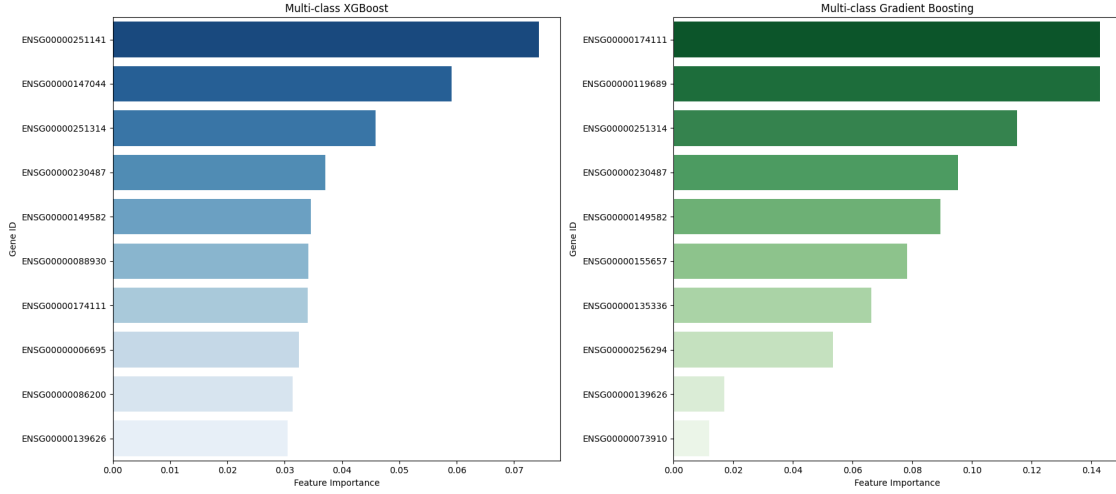


Figure 4: Top 10 most important features identified by the multiclass classification models: XGBoost and Gradient Boosting.

The results indicate a limited overlap in the exact top-ranked genes in the four classifiers, reflecting differences in the way each algorithm evaluates the importance of the features. Logistic Regression identifies features based on coefficient magnitude, whereas tree-based models such as Random Forest, XGBoost, and Gradient Boosting rank features according to their contribution to decision splits.

When compared with the biomarker genes reported in Best et al. (2015), minimal direct overlap was observed. However, this does not necessarily reduce the relevance of the identified biomarkers, as different machine learning algorithms often detect alternative but equally informative transcriptomic signatures depending on model architecture and feature selection strategy.

4.4 Robustness Analysis Using Noisy Data

As an optional robustness assessment, noise was added to the input transcriptomic data and feature importance analysis was repeated. The noisy-data analysis revealed that all models retained meaningful predictive structure; however, shifts in feature rankings were observed across both binary and multiclass classifiers. No overlap was observed between the noisy-data top-ranked genes and the biomarker list reported in Best et al. (2015), indicating that noise can substantially alter feature prioritisation.

These findings suggest that while predictive performance remains reasonably robust, biomarker ranking stability is sensitive to data quality. This highlights the importance

of preprocessing, noise control, and robustness validation in transcriptomic biomarker discovery.

4.5 Overall Discussion

The results demonstrate that machine learning methods can effectively classify platelet RNA sequencing data for cancer detection, particularly for binary classification tasks. Binary models achieved substantially stronger performance than multiclass models, reflecting the lower complexity of distinguishing healthy samples from cancer cases compared with separating multiple cancer subtypes with overlapping biological characteristics.

The weaker multiclass performance is likely due to several factors, including the increased complexity of distinguishing biologically similar cancer classes, the high dimensionality of transcriptomic data, class imbalance, and the relatively limited number of samples available for individual cancer subtypes. Further improvements could potentially be achieved through hyperparameter optimisation of the multiclass models, dimensionality reduction techniques such as Principal Component Analysis (PCA), more advanced feature selection methods, improved class balancing strategies, and exploration of alternative ensemble or deep learning approaches. Ensemble learning methods such as Random Forest, Gradient Boosting, and XGBoost proved effective for modelling nonlinear transcriptomic relationships, while Logistic Regression provided a strong and interpretable baseline model for binary classification.

Feature importance analysis revealed substantial variability between models, indicating that different algorithms identify different predictive biomarker signatures. Small overlap was observed among the models developed in this study, particularly ENSG00000155657 and ENSG00000086200, suggesting these transcripts represent robust predictive biomarkers across multiple machine learning approaches. This reinforces the importance of evaluating multiple machine learning approaches in transcriptomic biomarker discovery rather than relying on a single classification method.

Overall, the findings support the suitability of machine learning for platelet transcriptomics-based cancer classification. While the developed models achieved performance comparable to prior benchmark work for binary classification, further methodological refinement is required to improve multiclass cancer subtype prediction. ““

5 Contributions

- Bilal Ashraf: Developed and train at least two ML classifier (Binary) and wrote first half of report.
- Kompal Fayyaz: Developed and train multi class classifiers and contributed to result and discussion of report.